models of the Earth's circulating atmosphere. As an aid to understanding earthquakes, geophysicists build simple theoretical models of slipping rocks. As an aid to understanding stellar implosion, Oppenheimer and Snyder in 1939 built a simple theoretical model: an imploding cloud of matter that was perfectly spherical, perfectly homogeneous, and completely devoid of pressure. And as we physicists build all these models, we are intensely aware of their limitations. They are but pale images of the complexity that abounds "out there," in the "real" Universe.

Not so for a black hole—or, at least, not so once the radiation has flown away, carrying off all the hole's "hair." Then the hole is so exceedingly simple that we can describe it by precise, simple mathematical formulas. We need no idealizations at all. Nowhere else in the macroscopic world (that is, on scales larger than a subatomic particle) is this true. Nowhere else is our mathematics expected to be so precise. Nowhere else are we freed from the limitations of idealized models.

Why are black holes so different from all other objects in the macroscopic Universe? Why are they, and they alone, so elegantly simple? If I knew the answer, it would probably tell me something very deep about the nature of physical laws. But I don't know. Perhaps the next generation of physicists will figure it out.

Black Holes Spin and Pulsate

What are the properties of the hairless holes, which are described so perfectly by the mathematics of general relativity?

If a black hole is idealized as having absolutely no electric charge and no spin, then it is precisely the spherical hole that we met in previous chapters. It is described, mathematically? by Karl Schwarz-schild's 1916 solution to Einstein's field equation (Chapters 3 and 6).

When electric charge is dropped into such a hole, then the hole acquires just one new feature: electric field lines, which stick out of it radially like quills out of a hedgehog. If the charge is positive, then these electric field lines push protons away from the hole and attract electrons; if it is negative, then the field lines push electrons away and attract protons. Such a charge-endowed hole is described mathematically, with perfect precision, by a solution to Einstein's field equation found by the German and Dutch